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Modeling the interaction between vehicle yielding and pedestrian crossing behavior at unsignalized midblock crosswalks

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ABSTRACT

The interaction between pedestrians and vehicles is an inevitable phenomenon at unsignalized midblock crosswalks. This study aims to contribute to a better understanding of the interaction between vehicle yielding and pedestrian gap acceptance (VY and PGA). A microscopic traffic flow model was established to describe the interaction and explores its effect on traffic flow. The VY and PGA behaviors were converged into the proposed model. The proposed model was accomplished in a time step simulation. The results stability and descriptive power of the proposed model were analyzed. The proposed model was also validated using empirical data. The effects of the traffic and geometric factors on the operation of the unsignalized midblock crosswalks were discussed based on numerical experiments. Accordingly, the recommendations on choosing the proper control mode of midblock crosswalks (unsignalized or signalized) were proposed.

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1. Introduction

The interaction between motor vehicles and non-motorized traffic at crosswalks is a normal occurrence in urban centers as people move from one side of the road to the other. The economic and social activities bestriding the roads (such as stores, hospitals, hotels, and educational institutions), the composition of vehicle traffic (such as cars, buses, and trucks) and road geometry affect the frequency of these interactions. Where there is no signalized traffic control, these activities can generate intermittent pedestrian traffic crisscrossing the road segment and interfering with vehicle traffic flow.

Most people may not be willing to walk a longer distance to the signalized crosswalks at intersections if crossing the street where they are will give them convenient and direct access to their destination (Islam, Serhiyenko, Ivan, Ravishanker, & Garder, 2014; Kadali & Vedagiri, 2013; Zhao, Ma, & Li, 2017). Urban transportation planners use the midblock crosswalks to provide a convenient and orderly utilization of road facilities between the vehicles and pedestrians (Tezcan, Elmorssy, & Aksoy, 2019). To avoid a regular interruption on vehicular traffic flow, the unsignalized midblock crosswalks are widely used, thus potentially leading to conflicts (Yang, Liu, Xu, & Xu, 2016). Whether a midblock crosswalk is properly set and whether it should be controlled by the signal is a puzzle for regulatory agencies. It depends on the interaction between vehicles and pedestrians (Kutela & Teng, 2019). The three related topics, namely pedestrian gap acceptance, vehicle

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yielding behavior, and microscopic traffic flow model, have been separately studied. Although many microscopic traffic flow models have been established to analyze various complex traffic phenomena, the existing car-following models did not consider the interaction between vehicle yielding and pedestrian gap acceptance (VY and PGA) behaviors. The gap acceptance and yielding are the important decisions made by pedestrians and vehicles when passing the unsignalized midblock crosswalk. The motivation of the paper is to contribute to a better understanding of the vehicle-pedestrian interaction phenomenon at unsignalized midblock crosswalks by converging the probabilistic models of VY and PGA into the car-following model.

The paper is organized as follows. Section 2 reviews the relevant studies. Section 3 establishes the microscopic traffic flow model and the simulation program. Section 4 validates the results stability, descriptive power, and the accuracy of the proposed model. Section 5 analyzes the sensitivity of influencing factors. The conclusions are drawn in Section 6.

2. Literature review

This study explores the interaction between vehicles and pedestrians at unsignalized midblock crosswalks, where the two parties share sections of a roadway with or without conflicts in a traffic stream. The main actions that define this interaction are the driver slowing down and yielding (or failing to yield) on one side while the pedestrian waits or accepts the available gap and crosses. When crosswalks are unsignalized, decisions of whether a vehicle yields or pedestrians accept available gaps rests on the judgment of both parties at that point in time when they need to share the resource without conflicts (Zhang, Chen, & Wei, 2019). These decisions and subsequent actions are greatly influenced by the behavior of both parties. In the existing studies, there are three related topics, namely pedestrian gap acceptance, vehicle yielding behavior, and microscopic traffic flow model.

On the aspect of pedestrian gap acceptance, the behavior of pedestrians and how they perceive and respond to the evolving traffic ecosystem at the unsignalized midblock crosswalk may be influenced by their character and environmental factors (Chai, Shi, Wong, Er, & Gwee, 2016; Leden, Gårder, & Johansson, 2006; Papadimitriou, Yannis, & Golias, 2009; Wang, Tan, Schwebel, Shi, & Miao, 2018). Studies show that bold, assertive, and aggressive pedestrian crossing behavior as well as being in big platoons can make drivers who may be otherwise unwilling to give way to yield (Tasic, Elvik, & Brewer, 2017; Varhelyi, 1998). The composition of pedestrians waiting group (based on gender, age groups, disability status and whether they bear luggage) also affects their decision to either accept or reject a gap (Ferenchak, 2016). Reckless pedestrians tend to "comply" with the need to wait if in a large group of waiting pedestrians but a large group also can embolden them to take a risky gap believing in security in numbers (Elvik, 2016). Built-in safety features such as median island reservations, pedestrian refuges, safety guardrails, markings on the road surface and sidewalks and clear visibility for pedestrians and drivers in the area of the crossing, play a big part in the quality level of service for pedestrian facilities. An urban traffic ecosystem with multilane roads further presents a unique challenge to pedestrians. If they wait too long, then they may be impatient and choose the rolling gap (Petzoldt, 2014) as the most convenient (but risky if traffic speed is high) as they cross and stop one lane at a time. Zhang, Zhou, Chen, and Chen (2017) proposed an ordered probit model of pedestrian-vehicle conflict analysis to explore the effect of influencing factors, including traffic volume, vehicle speed, pedestrian crossing behavior, pedestrian refuge, on pedestrian-vehicle conflicts. A pedestrian dynamic gap acceptance model was further proposed to describe pedestrians' crossing behaviors at any multi-lane crosswalk. (Zhang, Zhou, Qiu, & Liu, 2018). Kadali, Vedagiri, and Rathi (2015) the present study adopts both the multiple linear regression and the artificial neural network with different pedestrian, traffic and vehicular characteristics to assess the significant contributing factors for pedestrians' gap acceptance behavior at unprotected mid-block crosswalks under mixed traffic conditions. It shows that pedestrian rolling behavior has a significant effect on pedestrian accepted gap size.

On the aspect of drivers, when they find pedestrians at the crosswalks, their behavior is affected by their character, type of vehicle, and the traffic ecosystem (Goddard, Kahn, & Adkins, 2015). This midblock crosswalk encounter is also characterized by subtle but non-verbal communication of intent between the two parties (Guéguen, Eyssartier, & Meineri, 2016). Both parties in traffic need to read and correctly perceive each other's actions (such as hand signals, flashing lights or eye contact) and intentions to be able to act accordingly (Obeid, Abkarian, Abou-Zeid, & Kaysi, 2017; Sucha, Dostal, & Risser, 2017; Zhuang & Wu, 2014). Merlino and Mondada (2019) explored the interaction in terms of different forms of mobility, distinct rights and obligations of pedestrians, the practices, and the visual resources such as gaze/eye-contact and gestures through which pedestrians execute the sequential and temporal trajectories of the crossing. According to existing studies, visible enforcement of traffic laws has been found to have a positive effect on driver yielding (Houten, Malenfant, Huitema, & Blomberg, 2013). Also, drivers approaching the crosswalks at higher speeds (Bertulis & Dulaski, 2018; Wang, Jiang, Wang, & Cheng, 2016) or in a platoon were less likely to give way to pedestrians (Hatfield, Fernandes, Job, & Smith, 2007). On the other hand, drivers are likely to be more careful in the presence of many pedestrians and, particularly, the vulnerable children and elderly. The conspicuity of a crosswalk can be enhanced by many measures, such as placing elevated signs (Costa, Bonetti, Vignali, Lantieri, & Simone, 2018; Costa et al., 2014), installing raised pedestrian crosswalks (Gitelman, Carmel, Pesahov, & Chen, 2017), setting the median refuge island and "Yield here to pedestrians" flashing vertical sign (Vignali et al., 2019), and using automatic pedestrian detection (Høye & Laureshyn, 2019). Bella, Natale, and Silvestri (2017) further analyzed drivers' behavior while approaching pedestrian crosswalks to assess the effectiveness of Advanced Driving Assistance Systems for pedestrian detection among several road environments.

Some other studies also focused on exploring the severity of conflicts. Lee and Abdel-Aty (2005) estimated the likelihood of pedestrian injury severity when pedestrians are involved in crashes using an ordered probit model based on the vehicle–pedestrian crashes at intersections in Florida over 4 years. They also developed a logical measure of exposure to better reflect pedestrian crash risk by using the information on individual walking trips in the household travel survey. Almodfer, Xiong, Fang, Kong, and Zheng (2016) proposed a lane-based approach for evaluating the post-encroachment time between pedestrians and vehicles. The lane-based distribution of pedestrian-vehicle conflict was analyzed. Vedagiri and Kadali (2016) explored the severity of pedestrian-vehicle conflicts at unsignalized midblock crosswalks under mixed traffic conditions by considering the pedestrian-vehicle conflicts based on a pedestrian safety margin defined as the time difference between the time gap of an approaching vehicle and the pedestrian crossing time. Ni, Wang, Sun, and Li (2016) explored the evaluation of pedestrians' safety with attention to the effects of behavioral factors. Three interaction patterns, namely hard interaction, no interaction, and soft-interaction were considered. However, none of these studies converged the interaction into the microscopic traffic flow model, which makes it difficult to conduct further study, such as analyzing the impact on traffic flow

On the aspect of the microscopic traffic flow model, it is a useful method to depict, reproduce, and predict the actual traffic phenomena by describing the dynamic characteristics of traffic motion (Gupta & Katiyar, 2006; Yu, Tang, & Xin, 2018; Zhu & Zhang, 2018). The car-following model is one of the most important parts in the field of microscopic traffic flow. It can be divided into the stimulus-response model, collision avoidance model, driving psychophysical model, and artificial intelligence-based model (Li, Jiang, & Jia, 2011). Among them, the optimal velocity model (Helbing & Tilch, 1998; Jiang, Wu, & Zhu, 2001; Nagatani, 1999; Zhao & Gao, 2005) has received extensive attention due to its analyzable and easy to simulate. Jiang et al. (2001) developed the full velocity difference (FVD) car-following model to address the problems of too high acceleration and unrealistic deceleration in the optimal velocity model (Bando et al., 1995) and GFM (Helbing & Tilch, 1998) by introducing positive relative velocity into the GF model to improve the traffic flow. Subsequently, other studies that have extended the FVD model considered the anticipation optimal velocity (Peng & Cheng, 2013), driver's anticipation behavior (Kang, Sun, & Yang, 2015; Tang, Li, & Huang, 2010), timid and aggressive driving behaviors (Peng, He, & Lu, 2016), multiple information of preceding cars (Peng & Sun, 2010), roadside memorial (Tang, Wu, Caccetta, & Huang, 2011), speed guidance at intersections (Zhao & Li, 2016), traffic jerk effect (Cheng, Liu, & Ge, 2017) and lately, illegal pedestrians at midblock crosswalks (Malenje, Zhao, Li, & Han, 2018). For the vehicle-pedestrian interaction, Pretto, Cybis, and Jacobsen (2011) presented a pedestrian road-crossing model, which focused on the representation of the pedestrians' behavior by combining force field and rule based approaches. Daganzo and Knoop (2016) explored the effect of the pedestrianized street on car traffic. The results show that the street's capacity is inversely proportional to the square root of the pedestrian flux for low pedestrian fluxes, and that pedestrians increase the cars' free-flow pace by an amount that is proportional to the pedestrian flux. Recently, Yang, Kurt, Redmill, and Özgüner (2017) presented a vehicle-pedestrian interaction simulator based on social force model for intelligent vehicles to achieve better decision making and local path planning. The simulator can effectively reflect the interaction behavior between vehicles and pedestrians. They proposed two pedestrian trajectory datasets, CITR dataset and DUT dataset, which can be used to calibrate and verify the pedestrian motion models with the consideration of vehicular influences (Yang, Li, Redmill, & Ozguner, 2019).

The three topics related to this problem, including pedestrian gap acceptance, vehicle yielding, and car-following, have been studied and analyzed but separately. Although lots of works have been conducted to investigate the decision-making process of pedestrians and vehicles, we still cannot describe a full picture of the vehicle-pedestrian interaction phenomenon at unsignalized midblock crosswalks. Therefore, this study converged the probabilistic models of VY and PGA into the car-following model. Vehicles can react according to the gap acceptance decision of pedestrians, and vice versa. In the way, the traffic operation at unsignalized midblock crosswalks can be simulated with the consideration of the interaction between vehicles and pedestrians.

3. Modeling

In this section, we extend the traditional FVD model with the consideration of the interaction between VY and PGA behavior at midblock crosswalks. The proposed car-following model integrates the judgment behaviors of the drivers and pedestrians.

3.1. Traffic situation

In vehicle-pedestrian interactions at the unsignalized midblock crosswalk, drivers respond to the presence of pedestrians by slowing down to create a gap that allows them to cross and pedestrians respond by either accepting or rejecting the gap. Pedestrians can only cross if there is a suitable gap to cross without the risk of being knocked down by a speeding oncoming vehicle. A suitable gap is the one where pedestrians can commence and complete crossing before the car in front of the traffic stream reaches the stop line of the crosswalk. Therefore, drivers approaching the crosswalk need to slow down (or yield) to avoid potential conflicts with pedestrians. If there is a stream of crossing pedestrians, they not only slow down but have to stop if the number of pedestrians is large. However, there is a threshold beyond which no pedestrian is expected to commence crossing if they have not started crossing when the vehicle arrives at the crosswalk.

The behavior of drivers and pedestrians is expected to be stochastic and depends on the conditions at the crosswalks at a particular time. In the model, each vehicle should decide whether to yield according to the logit model when it faces a crosswalk (no vehicle exists between the crosswalk and the target vehicle). When there is no pedestrian or pedestrian is crossing, the driver has no choice but to proceed and yield, respectively. Similarly, the pedestrians should decide whether to cross according to the gap acceptance logit model when they face a vehicle. When a gap is rejected, the pedestrian should wait until the vehicle has passed the crosswalk, then respond to the new gap available. When the vehicle yields forwardly or anyone in the waiting crowds decide to cross, the decision will be remade. The interaction will be defined by the interaction of the two logit models in the car-following model according to the conditions in Fig. 1.

Generally, the car following model is a microscopic traffic flow model that describes a typical urban traffic flow based on the behaviors of individual drivers. It can be described generally by the following expression.

$$\frac{dv_n(t)}{dt} = f(v_n, \Delta x_n, \Delta v_n) \tag{1}$$

where v_n, Δx_n, and Δv_n are the nth vehicle's velocity, headway, and relative velocity respectively, $f(v_n, \Delta x_n, \Delta v_n)$ is the stimulus function determined by the nth vehicle's velocity, headway, relative velocity and other factors.

If the vehicle has passed the crosswalk or the ahead vehicle has not passed the crosswalk, or the vehicle decides not to yield, the vehicle should continue without stopping according to the traditional FVD car-following model (Helbing & Tilch, 1998; Jiang et al., 2001), as shown in Eq. (2).

$$\frac{dv_n(t)}{dt} = \kappa[V(\Delta x_n(t)) - v_n(t)] + \lambda \Delta v_n(t), (x_n(t) \geq x_c or x_{n-1}(t) < x_c or y_n(t) = 0) \tag{2}$$

where $V(.)$ is the optimal velocity function, in m/s, which can be calculated by Eq. (3); $x_n(t)$ is the position of vehicle n at time t; $\Delta x_n(t)$ is the distance between vehicle n and vehicle $n-1$ at time t, in m, as shown in Eq. (4); $v_n(t)$ is the velocity of vehicle n at time t, in m/s; $\Delta v_n(t)$ is the velocity difference between vehicle n and vehicle $n-1$ at time t, in m/s, as shown in Eq. (5); $y_n(t)$ is the decision of the vehicle n whether to yield at time t, 1-yield, 0-not yield; x_c is the position of the crosswalk, in m; κ and λ are sensitivity parameters.

$$V(\Delta x_n(t)) = V_1 + V_2 \tanh(C_1(\Delta x_n(t) - L_c) - C_2) \tag{3}$$

where L_c is the length of the vehicle, in m; V_1, V_2, C_1, C_2 are parameters.

$$\Delta x_n(t) = x_{n-1}(t) - x_n(t) \tag{4}$$

$$\Delta v_n(t) = v_{n-1}(t) - v_n(t) \tag{5}$$

If the vehicle decides to yield, the vehicle will slow down by treating the crosswalk as the stop line and then stopping if the pedestrians take longer to cross, as shown in Eq. (6).

$$\frac{dv_n(t)}{dt} = \kappa[V(d_n(t)) - v_n(t)] - \lambda v_n(t), (x_n(t) < x_c \ \ and \ \ x_{n-1}(t) \geq x_c \ \ and \ \ y_n(t) = 1) \tag{6}$$

where $d_n(t)$ is the distance between vehicle n and the crosswalk at time t, in m, as shown in Eq. (7).

$$d_n(t) = x_c - x_n(t) \tag{7}$$

It is important to note whether a driver decides to yield or not depends on two things: the existence of pedestrians crossing on the crosswalk and the willingness of the drivers. If pedestrians are crossing, the driver has to yield. If there is no waiting pedestrian, the driver does not need to yield. If there are pedestrians who are waiting but not crossing, the driver should decide to yield or not. It can be specified as Eq. (8).

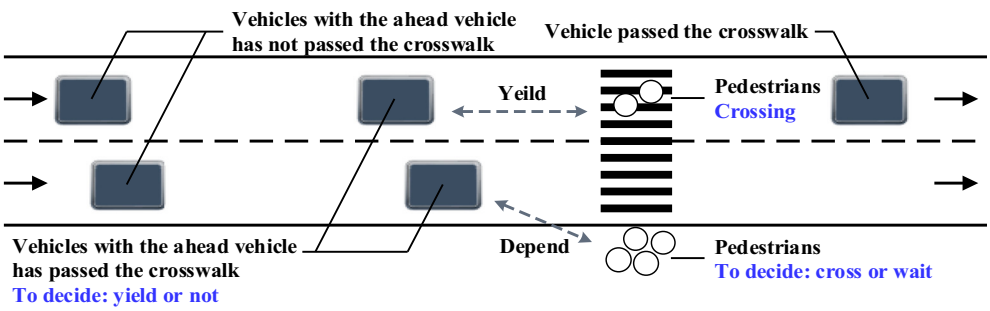


Fig. 1. Traffic flow with vehicle-pedestrian interaction.

$$y_n(t) = \begin{cases} 1, & c(t) = 1 \\ \lfloor \frac{p_n^y(t)}{\delta_n} \rfloor, & c(t) = 0 \ and \ N_p(t) > 0 \\ 0, & c(t) = 0 \ and \ N_p(t) = 0 \end{cases} \tag{8}$$

where $c(t)$ is the existence of pedestrians crossing on the crosswalk at time t, 1-yes, 0-no; $N_p(t)$ is the number of waiting pedestrians, in ped;$p_n^y(t)$ is the logistic probability of the yielding of vehicle n at time t; δ_n is the threshold of the yielding of vehicle n.

3.2. Pedestrian crossing and vehicle yielding behavior

The interaction between pedestrians and vehicles is a convergence between the gap acceptance and yielding behaviors at the crosswalks.

The pedestrian gap acceptance behavior is guided by the judgment of the pedestrians according to the following four factors: gap size, crossing distance, pedestrians waiting time, and location of waiting pedestrians (curbside or median). These factors are combined in the following logistic model established by Zhao, Malenje, Tang, and Han (2019) based on the data collected from 13 midblock crosswalk locations in Shanghai, China, as shown in Eq. (9). The utility function for the gap acceptance of pedestrians, as shown in Eq. (10), can be replaced by any suitable models without changing the framework of the proposed model, e.g., as the vehicle waiting time at the crosswalk increases, drivers are more likely to accept a shorter gap between pedestrians to reduce delay. It can be added to Eq. (10) as an influencing factor.

$$p_{il}^g(t) = \frac{1}{1 + e^{-U_{il}(t)}} \tag{9}$$

where $p_{il}^g(t)$ is the probability of the accepting gap on lane l for the pedestrian i; U_{il} is the multiple linear regression function for the accepting gap, as shown in Eq. (10) (Zhao et al., 2019).

$$U_{il}(t) = 6.365 + 2.678G_l(t) - 2.846D + 0.058W_i(t) - 1.273S_i(t) \tag{10}$$

where $G_l(t)$ is the gap size on lane l at time t, in s; D is the length of the crosswalk, in m; $W_i(t)$ is the waiting time, in s; $S_i(t)$ is the position of a pedestrian, 1-kerbside and 0-median.

The variable $p_{il}^g(t)$ is a continuous variable, which indicates the possibility of the gap acceptance, but the gap acceptance is binary (accept or reject gap). Therefore, the decision can be made by comparing the logistical probability $p_{il}^g(t)$ and the threshold of the gap acceptance of pedestrian i. It is important to note that under the multi-lane condition, the pedestrian will decide to cross only when the gaps on all the traffic lanes in one direction are accepted, as shown in Eq. (11).

$$g_i(t) = \lfloor \frac{\min(p_{il}^g(t))}{\eta_i} \rfloor \tag{11}$$

where $g_i(t)$ is the decision of the pedestrian i whether to cross at time t, 1-cross, 0-wait; $p_{il}^g(t)$ is the logistic probability of the crossing of pedestrian i at time t; η_i is the threshold of the crossing of pedestrian i; $\lfloor . \rfloor$ is the floor function.

The yielding behavior of the vehicle is guided by the judgment of the driver according to the following factors: gap size, number of lanes, number of waiting pedestrians, location of waiting pedestrians (curbside or median), and presence (or absence) of e-police. These factors are combined in a logistic model established by Malenje, Zhao, Li, and Han (2019), as shown in Eq. (12). The utility function for the yielding behavior of vehicles, as shown in Eq. (13), can be replaced by any suitable models without changing the framework of the proposed model.

$$p_n^y(t) = \frac{1}{1 + e^{-Z_n(t)}} \tag{12}$$

where $Z_n(t)$ is the multiple linear regression function for the yielding, as shown in Eq. (13) (Malenje et al., 2019).

$$Z_n(t) = 1.291 + 2.937G_n(t) - 8.462N_l + 0.730N_p(t) + 3.008S(t) + 11.913E_P \tag{13}$$

where $G_n(t)$ is the gap size for vehicle n at time t, in s; N_l is the number of lanes; $S(t)$ is the existence of the pedestrian waiting at the median, 1-yes and 0-no; E_P is the presence of the e-police, 1-yes and 0-no.

3.3. Simulation

The proposed model is realized by a time step simulation in MATLAB(R2018a) to describe the interaction between VY and PGA behaviors at unsignalized midblock crosswalks. The process of the simulation is shown as follows.

Step 1: Input data, including the position of the crosswalk (x_c), the length of the crosswalk (D), the number of traffic lanes (N_l), the existence of the median island (M), the presence of the e-police (E_P), the arrival distributions of vehicles and pedestrians, the distribution of the threshold of the gap acceptance for pedestrians, the distribution of the walking speed for pedestrians, the distribution of the threshold of the yielding for vehicles, and the parameters used in the car-following model (κ, λ, V_1, V_2, C_1, C_2, l_c).

Step 2: Generate new vehicles and pedestrians at the entrances according to the arrival distributions of vehicles and pedestrians. Initialize the threshold of the yielding (δ_n) for the newly generated vehicles. Initialize the threshold of the gap acceptance (η_i) and the walking speed (v_i) for the newly generated pedestrians.

Step 3: Determining the behavior of pedestrians. For any pedestrian i who are crossing ($i \in \mathcal{C}(t)$), make a judgment whether the crossing has been finished, as shown in Eq. (14). For any pedestrian i who are waiting ($i \in \mathcal{W}(t)$), make a judgment whether the gap is accepted using Eq. (11). If the gap is accepted, the pedestrian is going to cross, as shown in Eq. (15). Then, the state of the crosswalk (the existence of pedestrians crossing on the crosswalk) can be determined by Eq. (16).

$$c_i(t) = \begin{cases} 1, T_i(t) < \frac{D}{v_{pi}}, i \in \mathcal{C}(t) \\ 0, T_i(t) \geq \frac{D}{v_{pi}}, i \in \mathcal{C}(t) \end{cases} \tag{14}$$

where $\mathcal{C}(t)$ is a set of crossing pedestrians at time t; $c_i(t)$ is the state of pedestrians i at time t, 1-crossing, 0-not crossing; $T_i(t)$ is the duration of the crossing time of pedestrians i at time t, in s; v_{pi} is the walking speed of pedestrian i, in m/s. The distribution of the walking speed of pedestrians can be considered by setting the different walking speed for a different pedestrian.

$$c_i(t) = \begin{cases} 1, g_i(t) = 1, i \in \mathcal{W}(t) \\ 0, g_i(t) = 0, i \in \mathcal{W}(t) \end{cases} \tag{15}$$

where $\mathcal{W}(t)$ is a set of waiting pedestrians at time t.

$$c(t) = 1 - \prod_{i \in \mathcal{C} \cup \mathcal{W}} (1 - c_i(t)) \tag{16}$$

Step 4: Updating the state of pedestrians. Update the state of the pedestrian, as shown in Eqs. (17) and (18). Update the duration of the crossing time, as shown in Eq. (19). Update the waiting time, as shown in Eq. (20). Update the number of waiting pedestrians, as shown in Eq. (21).

$$\mathcal{W}(t + \Delta t) = \begin{cases} W(t) - \{i\}, g_i(t) = 1 \\ W(t), g_i(t) = 0 \end{cases} \tag{17}$$

where Δt is the length of the simulation time step, in s.

$$\mathcal{C}(t + \Delta t) = \begin{cases} C(t) \cup \{i\}, g_i(t) = 1 \\ C(t), g_i(t) = 0 \end{cases} \tag{18}$$

$$T_i(t + \Delta t) = T_i(t) + \Delta t, i \in \mathcal{C}(t) \tag{19}$$

$$W_i(t + \Delta t) = W_i(t) + \Delta t, i \in \mathcal{W} \tag{20}$$

$$N_p(t + \Delta t) = \text{card}(\mathcal{W}(t + \Delta t)) \tag{21}$$

where card(.) is the function calculating the number of elements in a set.

Step 5: Determining the behavior of vehicles. The yielding behavior can be determined by Eq. (8). Then, the vehicle can move according to the car-following model, as shown in Eqs. (2) and (6).

Step 6: Updating the state of vehicles. The Euler forward difference is selected to calculate the speed and position of each vehicle, as shown in Eqs. (22) and (23). Recording the entering time and the exiting time when a vehicle reaches the boundaries of the analysis segment [x_{b1}, x_{b2}], as shown in Eqs. (24) and (25).

$$v_n(t + \Delta t) = v_n(t) + \Delta t \frac{dv(t)}{dt} \tag{22}$$

$$x_n(t + \Delta t) = x_n(t) + v_n(t)\Delta t + \frac{1}{2}\frac{dv(t)}{dt}(\Delta t)^2 \tag{23}$$

$$t_{nb1} = t + \Delta t, x_n(t) < x_{b1} \ \ and \ \ x_n(t + \Delta t) \geq x_{b1} \tag{24}$$

$$t_{nb2} = t + \Delta t, x_n(t) < x_{b2} \ \ and \ \ x_n(t + \Delta t) \geq x_{b2} \tag{25}$$

where t_{nb1} and t_{nb2} are the time of a vehicle entering and the exiting the boundaries of the analysis segment, respectively, in m; x_{b1} and x_{b2} the two boundaries of the analysis segment, in m.

Step 7: Stopping criteria. The iteration will be stopped if the time step reaches the end of the analysis period (T), and output the indicators, including the average vehicular delay (the actual travel time minus the travel time under the desired speed) and the average pedestrian waiting time, as shown in Eqs. (26) and (27), respectively. Otherwise, update the time ($t = t + \Delta t$), then turn back to step 2.

$$D_{veh} = \frac{\sum (t_{nb2} - t_{nb1})}{\sum n} \tag{26}$$

$$W_{ped} = \frac{\sum W_i}{\sum i} \tag{27}$$

where D_{veh} is the average vehicular delay, in s; W_{ped} is the average pedestrian waiting time, in s.

In summary, the overall structure of the simulation framework is shown in Table 1 in a pseudo-code format.

4. Model validation

The results stability, descriptive power, and the accuracy of the proposed model are validated in this section. Firstly, the effect of the length of the analysis period on the stability of the calculation results is validated to make a recommendation on the length of the analysis period. Then, we determine whether the common interaction between vehicles and pedestrians can be reflected by the simulation. Last, the accuracy of the results of the simulation is validated using empirical data.

The parameters used in this section are set as follows. The number of traffic lanes is two. The start and end of the analysis area are between 0 m and 500 m. The unsignalized midblock crosswalk is located at 300 m. The length of the crosswalk equals 3.5 m multiply by the number of traffic lanes. The discretized time step is 0.1 s. The average arrival rate of vehicles and pedestrians are 250 veh/h/ln and 300 ped/h, respectively, with Poission distribution. The threshold of the gap acceptance for pedestrians and the threshold of the yielding for vehicles are set to be 0.5. The average walking speed for pedestrians is 1 m/s within the range of 0.5 m/s and 1.5 m/s. The parameters of the car-following model, including κ, λ, V_1, V_2, C_1, C_2, and l_c, are set to be 0.41, 0.2, 6.75, 7.91, 0.13, 1.57, and 9, respectively (Tang, Chen, Yang, & Shang, 2015; Tang et al., 2011; Zhao & Li, 2016).

4.1. Stability of the calculation results

The stability of the calculation results is affected by the length of the analysis period, as shown in Fig. 2. The model ran ten times for each given analysis period. It can be found that there was less dispersion in the average value under a given analysis period with the increase of the analysis length. Moreover, the standard deviation is relatively small (0.181 and 0.216 for the standard deviation of the waiting time of pedestrians and vehicles, respectively) when the length of the analysis period reaches to 7200 s (2 h).

4.2. Descriptive power validation

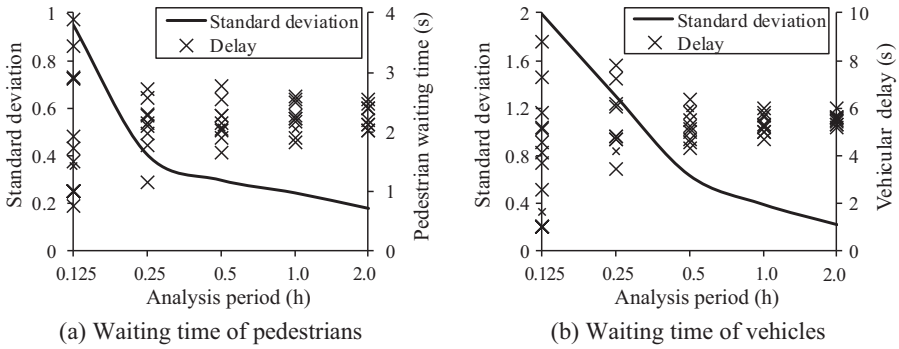
Fig. 3 shows an instance of the number of dwelling vehicles changing in each time-step. The two different colors of the running trail in Fig. 3(a) indicate the running trail of vehicles on different traffic lanes. In this period, 15 pedestrians arrive, 10 of them pass the crosswalk directly while the other five pedestrians wait a while before crossing. For vehicles, the vehicular delay depends on the remaining crossing time of the pedestrian when the vehicle arrives, which is related to the pedestrian walking speed, the length of the crosswalk, and the position of the pedestrian when the vehicle arrives. As shown in Fig. 3(b), for pedestrian 1, the corresponding vehicle does not yield because the vehicle is very close to the crosswalk. The pedestrian only waits 1.6 s. For pedestrians 2 and 3, they have to wait because the first two vehicles they met do not yield. However, the following vehicle yields, and then they can cross. The subsequent vehicles have to queue and wait until they finish the crossing. For pedestrian 5, he rejects the gap he meets. However, the vehicle decides to yield him. So, after waiting a short time, the pedestrian notices the vehicle yield (the vehicle slows down and the gap increases) and then starts to cross. It shows that the proposed model can describe the interaction between VY and PGA behaviors at unsignalized midblock crosswalks.

4.3. Empirical data validation

The proposed model converged the probabilistic models of VY and PGA into the car-following model, which were developed separately in previous studies (Malenje et al., 2019; Zhao et al., 2019). However, the accuracy of the separate behavioral models may not lead to the accuracy of the combined model. Therefore, in this section, the proposed model is validated using empirical data. The data were collected from two unsignalized midblock crosswalks in Shanghai, China, in a peak hour, as shown in Fig. 4. Since the previous studies covered the study area, the same parameter values for VY and PGA were used without recalibration. The arrival time and the time of starting to cross of pedestrians were recorded. Then, the average waiting time of pedestrians can be measured as the indicator to evaluate the veridicality of the model. The average value of the ten runs with 7200 s for each run is used as the result of simulation for comparison. The comparison results are shown in Table 2. One can find that the maximum relative error is less than 10%. Moreover, the Two-Related-Samples nonparametric test (Wilcoxon signed-rank) shows that there is no significant difference ($p > 0.05$) between the estimated and the surveyed values, as shown in Table 3. It indicates that the ability of the proposed model to estimate the operational performance at the unsignalized midblock crosswalk is acceptable.

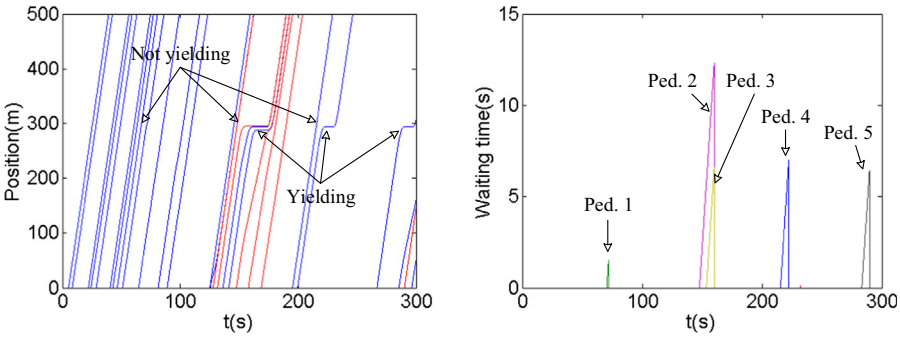
Table 1
Pseudo-code of the simulation.

```
Input data
for t = 1 to T
    Generate new vehicles and pedestrians
        if a new vehicle arrives
            Initialize the vehicle;
        end
        if a new pedestrian arrives
            Initialize the pedestrian;
        end
    Determine the behavior of pedestrians
        for i = 1 to NP (all the pedestrians)
            Calculate c_i(t) by Eqs. (14) and (15);
        end
        Calculate c(t) by Eq. (16);
    Update the state of pedestrians
        Update W(t + Δt), C(t + Δt), T_i(t + Δt), W_i(t + Δt), N_p(t + Δt) by Eqs. (17)-(21), respectively;
    Determining the behavior of vehicles
        Calculate y_n(t) by Eq. (8);
        if y_n(t) == 0
            Calculate dv_n(t)/dt by Eq. (2);
        else
            Calculate dv_n(t)/dt by Eq. (6);
        end
    Update the state of vehicles
        Calculate v_n(t + Δt), x_n(t + Δt), t_nb1, and t_nb2 by Eqs. (22)-(25), respectively;
end
Output the indicators D_veh and W_ped;
```



(a) Waiting time of pedestrians (b) Waiting time of vehicles

Fig. 2. Stability analysis.



(a) Running trail of vehicles (b) Pedestrian waiting time

Fig. 3. Model validation analysis.

(a) 295 Zhenhua Rd. (b) 1565 Lingshi Rd.

Fig. 4. Surveyed spots in Shanghai, China.

5. Sensitivity analysis

In this section, we present the results of the test for the reliability of the extended model under a range of test conditions. Specifically, the effects of vehicular volume, pedestrians' volume, and the number of traffic lanes were tested. To obtain a stable result in acceptable simulation time, the analysis period and the time-step are set to be 7200 s and 0.1 s, respectively. Moreover, the simulation will run 10 times for each scenario tested below.

5.1. Effect of the vehicular volume

The volume of the vehicles measured in terms of the number of vehicle units per hour per lane (veh/h/ln) was changed from 100 to 600 at 100 intervals. Other parameters are are constant, which is the same levels as that in Section 4. Comparing

Table 2
Comparison between the model results and empirical data.

Crosswalk	Lanes	Vehicle volume (veh/h/ln)	Pedestrian volume (person/h)	Pedestrian waiting time		
				Model (s)	Survey (s)	Relative error
295 Zhenhua Rd.	2	313	251	4.052	4.398	7.87%
1565 Lingshi Rd.	2	406	172	4.910	4.638	5.86%

Table 3
Result of the Two-Related-Samples nonparametric test (Wilcoxon signed-rank).

Ranks				Test Statistics	
Item	Number of samples	Mean Rank	Sum of Ranks	Z	Sig. (2-tailed)
Negative Ranks	9[a]	10.33	93.00	−0.448	0.654
Positive Ranks	11[b]	10.64	117.00		
Ties	0[c]				
Total	20				

Note: a. Survey result < Model result, b. Survey result > Model result, and c. Survey result = Model result.

the two sub-figures in Fig. 5, one can find that the vehicular delay increases more rapidly than the pedestrian waiting time with the increase of the volume. It is because the arriving vehicles have to queue once the vehicle in front yields and cannot depart simultaneously after pedestrians complete crossing. Additionally, the queue clearance takes time as vehicles have to restart and move according to the car-following model. With the increase in volume, more vehicles have to wait in the queue. Consequently, the vehicular delay increases more rapidly with the increase of the volume between 400 and 600 (see Fig. 5(a)). However, as shown in Fig. 5(b), pedestrians can depart simultaneously once a gap is available. Therefore, although the increase of the vehicular volume decreases the available gap for crossing, the influence on the waiting time of the pedestrians is slight if the driver willing to yield.

For the given parameters (collected from the field survey in Shanghai), the vehicles tend to yield. Under this condition, the difference in delay between vehicles and pedestrians is great, especially when the vehicular volume is high. The signal control is recommended when the vehicular volume is larger than 480 veh/h by using the vehicular delay of 20 s as a threshold, which is the criteria of the LOS (level of service) B for vehicles at signalized intersections (TRB, 2010). Please note that in Fig. 5, the volume is constant for each scenario. However, under the condition of the traffic demand fluctuation, e.g., during different times of day, it is unsuitable for controlling the crosswalk by fixed signal timing. A signal control with pushbutton is an alternative scheme to increase flexibility (Kutela & Teng, 2020).

5.2. Effect of the pedestrian volume

The presence and number of pedestrians at an unsignalized crosswalk do affect the response from drivers and influence their decision to either yield or not. In this test, the inputs for the volume of the pedestrians measured as the cumulative number of pedestrians per hour (ped/h) was set from 100 to 800 at 100 intervals. Other parameters are constant, which is the same as that in Section 4. The result of this test shows that the delay of vehicles increases more rapidly with the increase in the volume of pedestrians, as shown in Fig. 6(a). Notably, the more the pedestrian volume is, the more time the crosswalk will be occupied by pedestrians. Then, vehicles have to remain in the queue for a longer time. Moreover, after

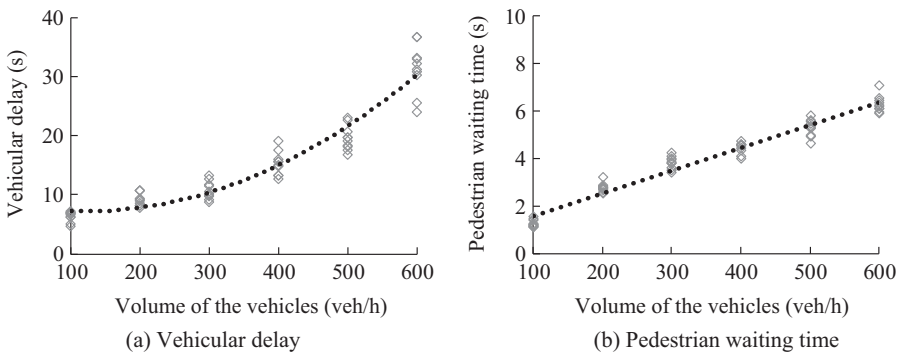


Fig. 5. Effect of the vehicular volume.

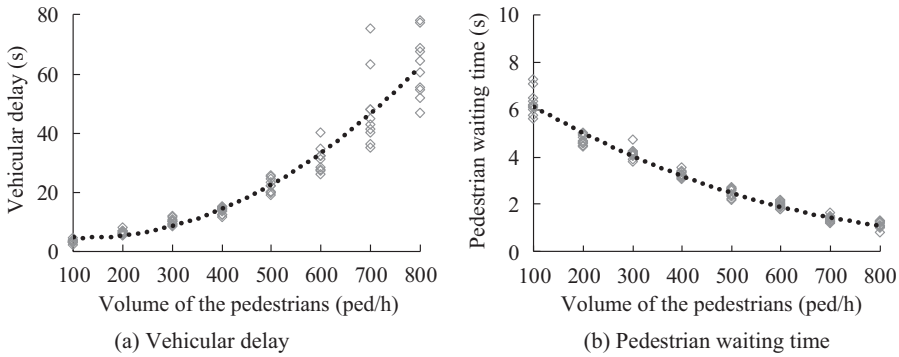


Fig. 6. Effect of the pedestrian volume.

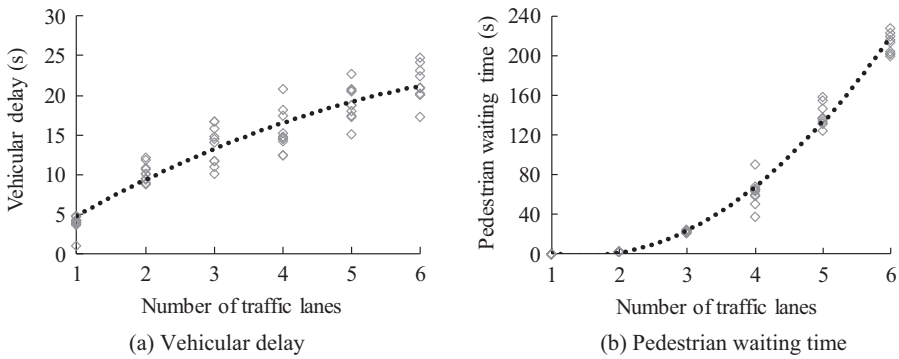


Fig. 7. Effect of the number of traffic lanes.

pedestrians complete crossing, vehicles have to depart in platoon and cannot depart simultaneously, which further enlarges the delay.

However, for pedestrians, the waiting time is inversely proportional to the volume of pedestrians. From pedestrians' perspective, with the increase in the pedestrian volume, more time is available for use by pedestrians, as shown in Fig. 6(b). It is because if one gap is accepted, the crosswalk will be occupied by the pedestrians for the duration of the crossing move, and in the process, vehicles have to slow down and yield to pedestrians to use the crosswalk. If the pedestrians arrive continuously, the crosswalk will be occupied for a long time. Therefore, along the same line as the vehicular volume analysis, the signal control is recommended when the pedestrian volume is larger than 500 ped/h by using the vehicular delay of 20 s as a threshold. E.g., the crosswalk can be controlled by the signal at the school area because the extreme peak crossing demand may occur during the morning and afternoon school time (Tang, Xie, Wang, & Zhang, 2019). It can also ensure the safety of the pedestrian.

5.3. Effect of the number of traffic lanes

The number of lanes affects the distance and duration that pedestrians need to traverse a crosswalk. In this test, the number of traffic lanes used was set to be 1, 2, 3, 4, 5, and 6, when other parameters are constant, which is the same as that in Section 4. Fig. 7 shows that, for the pedestrians, the increase in the number of traffic lanes makes the waiting time to increase rapidly. It is because of the sharp decline in the available gap. Moreover, the possibility that vehicles on all the traffic lanes yielding also drops substantially with the increase in the number of traffic lanes. Therefore, it is challenging for pedestrians to find a safe gap for crossing under the condition that the number of traffic lanes is high. According to Guo, Wang, Guo, Jiang, and Bubb (2012), most pedestrians can endure up to a 50 s wait. Therefore, it is not appropriate to have an unsignalized crosswalk set on a segment with more than 4 lanes. Instead, a central island should be added to reduce the number of traffic lanes the pedestrians have to cross in one time. Alternatively, the crosswalk can be signalized to clarify the right of way.

For the vehicles, the delay also increases with the increase in the number of traffic lanes. On average, 4 s increases in vehicular delay for one traffic lane increase. It is because the more the number of traffic lanes is, the longer time the pedestrians have to spend on the crossing.

6. Conclusion

This study explored the interaction of vehicles and pedestrians at the unsignalized midblock crosswalks, where the two parties share sections of a roadway with or without conflicts in a traffic stream based on the car-following theory. It involved developing a microscopic traffic flow model considering the behaviors of both the driver and pedestrians as represented in established logit models of VY and PGA. The numerical results show that the proposed model can describe the interaction and qualitatively analyze its effects on traffic flow. The operational performance of the unsignalized midblock crosswalks under different traffic and geometric conditions were explored based on the sensitivity analyses. The following are the key findings of the study.

(1) For pedestrians, the waiting time increases with the increase in the number of lanes and vehicular volume but decreases with the pedestrian volume. The number of lanes is the critical factor affecting the waiting time of the pedestrians at unsignalized midblock crosswalks.
(2) The vehicular delay increases with the growth of the vehicles' volume, the pedestrians' volume, and the number of lanes. Compared with the pedestrians, the vehicular delay increases more rapidly than the pedestrian waiting time with the increase of the vehicles' volume, pedestrians' volume.
(3) The signal control is recommended when the pedestrian waiting time and vehicular delay is high. It is related to the behavior of drivers and pedestrians. According to the parameters surveyed in this study, the signal control is recommended when the number of traffic lanes at the midblock crosswalk is more than 4.

This study explored the effect of vehicle-pedestrian interaction at a midblock unsignalized crosswalk where pedestrian crossing is legal though not controlled. The interaction under the condition the crosswalk is not set at the segment is worth to do further study in the future because the illegal crossing (cross at the point without the crosswalk) is typical in developing countries. It is a tradeoff whether a midblock crosswalk should be set. Moreover, the combined consideration of the midblock crosswalk and the adjacent intersections can be the direction of the future study. The arrival distribution of vehicles is strongly affected when the distance to adjacent intersections is considered.

CRediT authorship contribution statement

Jing Zhao: Conceptualization, Methodology, Software, Writing - review & editing. **Jairus Odawa Malenje:** Investigation, Writing - original draft, Methodology, Visualization. **Jingxian Wu:** Validation, Writing - review & editing. **Ruoming Ma:** Investigation, Visualization.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.trf.2020.06.019.

References

Almodfer, R., Xiong, S., Fang, Z., Kong, X., & Zheng, S. (2016). Quantitative analysis of lane-based pedestrian-vehicle conflict at a non-signalized marked crosswalk. *Transportation Research Part F: Traffic Psychology and Behaviour, 42*, 468–478.
Bando, M., Hasebe, K., Nakanishi, K., Nakayama, A., Shibata, A., & Sugiyama, Y. (1995). Phenomenological study of dynamical model of traffic flow. *Journal de Physique I, 5*, 1389–1399.
Bella, F., Natale, V., & Silvestri, M. (2017). Driver-pedestrian interaction under different road environments. *Transportation Research Procedia, 27* 148155.
Bertulis, T., & Dulaski, D. M. (2018). Driver approach speed and its impact on driver yielding to pedestrian behavior at unsignalized crosswalks. *Transportation Research Record, 2464*, 46–51.
Chai, C., Shi, X., Wong, Y. D., Er, M. J., & Gwee, E. T. M. (2016). Fuzzy logic-based observation and evaluation of pedestrians' behavioral patterns by age and gender. *Transportation Research Part F: Traffic Psychology and Behaviour, 40*, 104–118.
Cheng, R., Liu, F., & Ge, H. (2017). A new continuum model based on full velocity difference model considering traffic jerk effect. *Nonlinear Dynamics, 89*, 639–649.
Costa, M., Bonetti, L., Vignali, V., Lantieri, C., & Simone, A. (2018). The role of peripheral vision in vertical road sign identification and discrimination. *Ergonomics, 61*, 1619–1634.
Costa, M., Simone, A., Vignali, V., Lantieri, C., Bucchi, A., & Dondi, G. (2014). Looking behavior for vertical road signs. *Transportation Research Part F: Traffic Psychology and Behaviour, 23*, 147–155.
Daganzo, C. F., & Knoop, V. L. (2016). Traffic flow on pedestrianized streets. *Transportation Research Part B: Methodological, 86*, 211–222.
Elvik, R. (2016). Safety-in-numbers: Estimates based on a sample of pedestrian crossings in Norway. *Accident Analysis & Prevention, 91*, 175–182.
Ferenchak, N. N. (2016). Pedestrian age and gender in relation to crossing behavior at midblock crossings in India. *Journal of Traffic and Transportation Engineering (English Edition), 3*, 345–351.
Gitelman, V., Carmel, R., Pesahov, F., & Chen, S. (2017). Changes in road-user behaviors following the installation of raised pedestrian crosswalks combined with preceding speed humps, on urban arterials. *Transportation Research Part F: Traffic Psychology and Behaviour, 46*, 356–372.
Goddard, T., Kahn, K. B., & Adkins, A. (2015). Racial bias in driver yielding behavior at crosswalks. *Transportation Research Part F: Traffic Psychology and Behaviour, 33*, 1–6.

Guéguen, N., Eyssartier, C., & Meineri, S. (2016). A pedestrian's smile and drivers' behavior: When a smile increases careful driving. *Journal of Safety Research, 56*, 83–88.
Guo, H., Wang, W., Guo, W., Jiang, X., & Bubb, H. (2012). Reliability analysis of pedestrian safety crossing in urban traffic environment. *Safety Science, 50*, 968–973.
Gupta, A. K., & Katiyar, V. K. (2006). A new anisotropic continuum model for traffic flow. *Physica A: Statistical Mechanics and its Applications, 368*, 551–559.
Høye, A., & Laureshyn, A. (2019). SeeMe at the crosswalk: Before-after study of a pedestrian crosswalk warning system. *Transportation Research Part F: Traffic Psychology and Behaviour, 60*, 723–733.
Hatfield, J., Fernandes, R., Job, R. S., & Smith, K. (2007). Misunderstanding of right-of-way rules at various pedestrian crossing types: Observational study and survey. *Accident Analysis & Prevention, 39*, 833–842.
Helbing, D., & Tilch, B. (1998). Generalized force model of traffic dynamics. *Physical Review E, 58*, 133–138.
Houten, R. V., Malenfant, L., Huitema, B., & Blomberg, R. (2013). Effects of high-visibility enforcement on driver compliance with pedestrian yield right-of-way laws. *Transportation Research Record, 2393*, 41–49.
Islam, M. S., Serhiyenko, V., Ivan, J. N., Ravishanker, N., & Garder, P. E. (2014). Explaining pedestrian safety experience at urban and suburban street crossings considering observed conflicts and pedestrian counts. *Journal of Transportation Safety & Security, 6*, 335–355.
Jiang, R., Wu, Q., & Zhu, Z. (2001). Full velocity difference model for a car-following theory. *Physical Review E, 64* 017101.
Kadali, B. R., & Vedagiri, P. (2013). Modelling pedestrian road crossing behaviour under mixed traffic condition. *European Transport, 55*, 1–17.
Kadali, B. R., Vedagiri, P., & Rathi, N. (2015). Models for pedestrian gap acceptance behaviour analysis at unprotected mid-block crosswalks under mixed traffic conditions. *Transportation Research Part F: Traffic Psychology and Behaviour, 32*, 114–126.
Kang, Y., Sun, D., & Yang, S. (2015). A new car-following model considering driver's individual anticipation behavior. *Nonlinear Dynamics, 82*, 1293–1302.
Kutela, B., & Teng, H. (2020). Evaluating the influential factors for pushbutton utilization at signalized midblock crosswalks. *Safety Science, 122* 104533.
Kutela, B., & Teng, H. L. (2019). Modeling transitional states of drivers yielding right-of-way to pedestrians at signalized midblock crosswalks using a hazard-based multistate model. *Transportation Research Record, 2673*, 648–659.
Leden, L., Gårder, P., & Johansson, C. (2006). Safe pedestrian crossings for children and elderly. *Accident Analysis & Prevention, 38*, 289–294.
Lee, C., & Abdel-Aty, M. (2005). Comprehensive analysis of vehicle–pedestrian crashes at intersections in Florida. *Accident Analysis & Prevention, 37*, 775–786.
Li, L., Jiang, R., & Jia, B. (2011). *Modern traffic flow theory and application*. Beijing: Tsinghua University Press.
Malenje, J. O., Zhao, J., Li, P., & Han, Y. (2018). An extended car-following model with the consideration of the illegal pedestrian crossing. *Physica A: Statistical Mechanics and its Applications, 508*, 650–661.
Malenje, J. O., Zhao, J., Li, P., & Han, Y. (2019). Vehicle yielding probability estimation model at unsignalized midblock crosswalks in Shanghai. *China. PLoS one, 14* e0213876.
Merlino, S., & Mondada, L. (2019). Crossing the street: How pedestrians interact with cars. *Language & Communication, 65*, 131–147.
Nagatani, T. (1999). Stabilization and enhancement of traffic flow by the next-nearest-neighbor interaction. *Physical Review E, 60*, 6395–6401.
Ni, Y., Wang, M., Sun, J., & Li, K. (2016). Evaluation of pedestrian safety at intersections: A theoretical framework based on pedestrian-vehicle interaction patterns. *Accident Analysis & Prevention, 96*, 118–129.
Obeid, H., Abkarian, H., Abou-Zeid, M., & Kaysi, I. (2017). Analyzing driver-pedestrian interaction in a mixed-street environment using a driving simulator. *Accident Analysis & Prevention, 108*, 56–65.
Papadimitriou, E., Yannis, G., & Golias, J. (2009). A critical assessment of pedestrian behaviour models. *Transportation Research Part F: Traffic Psychology and Behaviour, 12*, 242–255.
Peng, G., & Cheng, R. (2013). A new car-following model with the consideration of anticipation optimal velocity. *Physica A: Statistical Mechanics and its Applications, 392*, 3563–3569.
Peng, G., He, H., & Lu, W. (2016). A new car-following model with the consideration of incorporating timid and aggressive driving behaviors. *Physica A: Statistical Mechanics and its Applications, 442*, 197–202.
Peng, G., & Sun, D. (2010). A dynamical model of car-following with the consideration of the multiple information of preceding cars. *Physics Letters A, 374*, 1694–1698.
Petzoldt, T. (2014). On the relationship between pedestrian gap acceptance and time to arrival estimates. *Accident Analysis & Prevention, 72*, 127–133.
Pretto, C. O., Cybis, H. B. B., Jacobsen, A. (2011) A multi-layer simulation model for vehicle and pedestrian interaction. Transportation Research Board 90th Annual Meeting, Washington DC, United States.
Sucha, M., Dostal, D., & Risser, R. (2017). Pedestrian-driver communication and decision strategies at marked crossings. *Accident Analysis & Prevention, 102*, 41–50.
Tang, T., Chen, L., Yang, S., & Shang, H. (2015). An extended car-following model with consideration of the electric vehicle's driving range. *Physica A: Statistical Mechanics and its Applications, 430*, 148–155.
Tang, T., Li, C., & Huang, H. (2010). A new car-following model with the consideration of the driver's forecast effect. *Physics Letters A, 374*, 3951–3956.
Tang, T., Wu, Y., Caccetta, L., & Huang, H. (2011). A new car-following model with consideration of roadside memorial. *Physics Letters A, 375*, 3845–3850.
Tang, T., Xie, C., Wang, T., & Zhang, J. (2019). Modeling and simulating the matching behavior of pedestrian flow at training school during the pickup period after class. *Physica A: Statistical Mechanics and its Applications, 521*, 649–660.
Tasic, I., Elvik, R., & Brewer, S. (2017). Exploring the safety in numbers effect for vulnerable road users on a macroscopic scale. *Accident Analysis & Prevention, 109*, 36–46.
Tezcan, H. O., Elmorssy, M., & Aksoy, G. (2019). Pedestrian crossing behavior at midblock crosswalks. *Journal of Safety Research, 71*, 49–57.
TRB (2010). *Highway capacity manual 2010*. Washington, DC: Transportation Research Board.
Varhelyi, A. (1998). Varhelyi Drivers' speed behaviour at a zebra crossing a case study. *Accident Analysis & Prevention, 30*, 731–743.
Vedagiri, P., & Kadali, B. R. (2016). Evaluation of pedestrian-vehicle conflict severity at unprotected midblock crosswalks in India. *Transportation Research Record, 2581*, 48–56.
Vignali, V., Cuppi, F., Acerra, E., Bichicchi, A., Lantieri, C., Simone, A., & Costa, M. (2019). Effects of median refuge island and flashing vertical sign on conspicuity and safety of unsignalized crosswalks. *Transportation Research Part F: Traffic Psychology and Behaviour, 60*, 427–439.
Wang, H., Tan, D., Schwebel, D. C., Shi, L., & Miao, L. (2018). Effect of age on children's pedestrian behaviour: Results from an observational study. *Transportation Research Part F: Traffic Psychology and Behaviour, 58*, 556–565.
Wang, T., Jiang, X., Wang, W., & Cheng, Q. (2016). Analysis of vehicle speed change at non-signalized crosswalks based on driving behavior. *Procedia Engineering, 137*, 547–553.
Yang, D., Kurt, A., Redmill, K., & Özgüner, Ü. (2017). Agent-based microscopic pedestrian interaction with intelligent vehicles in shared space. In *Proceedings of proceedings of the 2nd international workshop on science of smart city operations and platforms engineering* (pp. 69–74).
Yang, D., Li, L., Redmill, K., & Ozguner, U. (2019) Top-view trajectories: a pedestrian dataset of vehicle-crowd interaction from controlled experiments and crowded campus. In 2019 IEEE Intelligent Vehicles Symposium (IV), Paris, France (pp. 899–904).
Yang, Z., Liu, P., Xu, X., & Xu, C. C. (2016). Multiobjective evaluation of midblock crosswalks on urban streets based on TOPSIS and entropy methods. *Transportation Research Record*, 59–71.
Yu, S., Tang, J., & Xin, Q. (2018). Relative velocity difference model for the car-following theory. *Nonlinear Dynamics, 91*, 1415–1428.
Zhang, C., Chen, F., & Wei, Y. (2019). Evaluation of pedestrian crossing behavior and safety at uncontrolled mid-block crosswalks with different numbers of lanes in China. *Accident Analysis & Prevention, 123*, 263–273.
Zhang, C., Zhou, B., Chen, G., & Chen, F. (2017). Quantitative analysis of pedestrian safety at uncontrolled multi-lane mid-block crosswalks in China. *Accident Analysis & Prevention, 108*, 19–26.

Zhang, C., Zhou, B., Qiu, T., & Liu, S. (2018). Pedestrian crossing behaviors at uncontrolled multi -lane mid-block crosswalks in developing world. *Journal of Safety Research, 64*, 145–154.

Zhao, J., & Li, P. (2016). An extended car-following model with consideration of speed guidance at intersections. *Physica A: Statistical Mechanics and its Applications, 461*, 1–8.

Zhao, J., Ma, W., & Li, P. (2017). Optimal design of midblock crosswalk to achieve trade-off between vehicles and pedestrians. *Journal of transportation engineering, Part A: Systems, 143*, 04016003.

Zhao, J., Malenje, J. O., Tang, Y., & Han, Y. (2019). Gap acceptance probability model for pedestrians at unsignalized mid-block crosswalks based on logistic regression. *Accident Analysis & Prevention, 129*, 76–83.

Zhao, X., & Gao, Z. (2005). A new car-following model: Full velocity and acceleration difference model. *The European Physical Journal B-Condensed Matter and Complex Systems, 47*, 145–150.

Zhu, W. X., & Zhang, H. M. (2018). Analysis of mixed traffic flow with human-driving and autonomous cars based on car-following model. *Physica A: Statistical Mechanics and its Applications, 496*, 274–285.

Zhuang, X., & Wu, C. (2014). Pedestrian gestures increase driver yielding at uncontrolled mid-block road crossings. *Accident Analysis & Prevention, 70*, 235–244.